PRACTICE SHEET-1

1. If $\frac{2p}{p^2-2p+1}=\frac{1}{4}$, then the valve of $p+\frac{1}{p}$ is
(a) 4 (b) 5
(c) 10 (d) 12

2. If $x^2 - 3x + 1 = 0$, then the value of $x+\frac{1}{x}$ is
(a) 0 (b) 1
(b) 2 (d) 3

3. If $\frac{x^2-x+1}{x^2+x+1}=\frac{2}{3}$ then the value of $x+\frac{1}{x}$ is
(a) 4 (b) 5
(c) 6 (d) 8

4. If $2x+\frac{1}{3x}=5$, find the value of $\frac{5x}{6x^2+20x+1}$
(a) 1/4 (b) 1/6
(c) 1/5 (d) 1/7

5. If $\frac{x}{2x^2+5x+2}=\frac{1}{6}$, then value of $x+\frac{1}{x}$ is
(a) 2 (b) 1/2
(c) −1/2 (d) −2

6. If for non-zero x , $x^2 - 4x - 1 = 0$, then the value of $x^2+\frac{1}{x^2}$ is
(a) 4 (b) 10
(c) 12 (d) 18

7. If $x = 3 + \sqrt{8}$, then $x^2+\frac{1}{x^2}$ is equal to
(a) 38 (b) 36
(c) 34 (d) 30

8. If $x^2 - 4x - 1 = 0$, then the value of $x^2+\frac{1}{x^2}+3x-\frac{3}{x}$ is
(a) 0 (b) 10
(b) 20 (d) 30

9. If $x-\frac{1}{x}=4$, then $x+\frac{1}{x}$ is equal to
(a) $5\sqrt{2}$ (b) $2\sqrt{5}$
(c) $4\sqrt{2}$ (d) $4\sqrt{5}$

10. If $x = 5 + 2\sqrt{6}$, then the value of $\sqrt{x}+\frac{1}{\sqrt{x}}$ is
(a) $2\sqrt{2}$ (b) $3\sqrt{2}$
(c) $2\sqrt{3}$ (d) $3\sqrt{3}$

11. For a > b , If a + b = 5 and ab = 6, then the value of $a^2 - b^2$ is
(a) 1 (b) 3
(c) 5 (d) 7

12. If $x-\frac{1}{x}=5$, then $x^2+\frac{1}{x^2}$ is equal to
(a) 5 (b) 25
(c) 27 (d) 23

13. If $x = 3 + 2\sqrt{2}$, then $\sqrt{x}-\frac{1}{\sqrt{x}}$ is equal to
(a) 1 (b) 2
(c) $2\sqrt{2}$ (d) $3\sqrt{3}$

14. If $x=\sqrt{3}+\sqrt{2}$, then $x^2+\frac{1}{x^2}$ is equal to
(a) 4 (b) 6
(c) 9 (d) 10

15. If $x+\frac{9}{x}=6$, then $x^2+\frac{9}{x^2}$ is equal to
(a) 8 (b) 9
(c) 10 (d) 12

16. If $m+\frac{1}{m-2}=4$, then $(m-2)^2+\frac{1}{(m-2)^2}$ is equal to
(a) −2 (b) 0
(c) 2 (d) 4

17. If $x^2 + x + 1 = 0$, then the value of $x^3 + 1$ is
(a) 0 (b) 1
(c) 2 (d) 3

18. If $x+\frac{1}{x}=1$, then $x^{17}+\frac{1}{x^{17}}$ is equal to
(a) 0 (b) 1
(c) 2 (d) 3

19. If $x+\frac{1}{x}=1$, then $x^{16}+x^{13}$ is equal to

(a) 0 (b) 1
(c) 2 (d) 3

20. If $x+\frac{1}{x}=1$, then $x^{91}+x^{90}+x^{89}+x^{88}+x^{87}+x^{86}$ is equal to
(a) 0 (b) 1
(c) 2 (d) 3

21. If $x+\frac{1}{x}=\sqrt{3}$, then
$x^{93}+x^{91}+x^{87}+x^{85}+x^{83}+x^{89}$ is equal to
(a) 0 (b) 1
(c) 2 (d) 3

22. If $x^2+x+1=0$, then
$x^{93}+x^5+x^4+1=?$ is equal to
(a) 0 (b) 1
(c) 2 (d) 3

23. If $x+\frac{1}{x}=\sqrt{3}$, then
$x^{30}+x^{24}+x^{18}+x^{12}+x^6$ is equal to
(a) −3 (b) 1
(c) −1 (d) 2

24. What is the value of $\left[\frac{(2.3)^3-0.027}{(2.3)^2+0.69+0.09}\right]$
(a) 2.6 (b) 2
(c) 1.3 (d) 1

25. The value of

$$\frac{0.1\times0.1\times0.1+0.2\times0.2\times0.2+0.3\times0.3\times0.3-3\times0.1\times0.2\times0.3}{0.1\times0.1+0.2\times0.2+0.3\times0.3-0.1\times0.2-0.2\times0.3-0.3\times0.1}$$
is
(a) 0.006 (b) 0.6
(c) 0 (d) 0.2

26. The value of
$$\frac{0.9\times0.9\times0.9+0.2\times0.2\times0.2+0.3\times0.3\times0.3-3\times0.9\times0.2\times0.3}{0.9\times0.9+0.2\times0.2+0.3\times0.3-0.9\times0.2-0.2\times0.3-0.3\times0.9}$$
is
(a) 1.4 (b) 0.054
(c) 0.8 (d) 1.0

27. The value of
$\frac{(2.697-0.498)^2+(2.697+0.498)^2}{2.697\times2.697+0.498\times0.498}$ is
(a) 4 (b) 2
(c) 2.199 (d) 3.195

28. The value of
$\frac{0.0347\times0.0347\times0.0347+(0.9653)^3}{(0.0347)^2-0.0347\times0.9653+(0.9653)^2}$ is
(a) 0.9306 (b) 1.0009
(c) 1.0050 (d) 1

29. The value of
$\frac{(3.2)^3-0.008}{(3.2)^2+0.64+0.04}$ is
(a) 0 (b) 2.994
(c) 3.208 (d) 3

30. (113^2 + 115^2 + 117^2 −113 × 115 − 115 × 117 − 117 × 113) is equal to
(a) 0 (b) 4
(c) 8 (d) 12

ANSWER KEY																			
1.	(c)	2.	(d)	3.	(b)	4.	(d)	5.	(b)	6.	(d)	7.	(c)	8.	(d)	9.	(b)	10.	(c)
11.	(c)	12.	(c)	13.	(b)	14.	(d)	15.	(c)	16.	(c)	17.	(c)	18.	(b)	19.	(a)	20.	(a)
21.	(a)	22.	(a)	23.	(c)	24.	(b)	25.	(b)	26.	(a)	27.	(b)	28.	(d)	29.	(d)	30.	(d)

PRACTICE SHEET-2

1. If $\frac{a}{1-a}+\frac{b}{1-b}+\frac{c}{1-c}=1$ then the value of $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}$ is
(a) 1 (b) 2
(c) 3 (d) 4

2. If x + y =2z then the value of $\frac{x}{x-z}+\frac{z}{y-z}$ is
(a) 1 (b) 2
(c) 1/2 (d) 2

3. If $x^2 + y^2 + 2x + 1 = 0$, then the value of $x^{31} + y^{35}$ is
(a) −1 (b) 0
(c) 1 (d) 2

4. If a , b , c are real and $a^2 + b^2 + c^2 = 2(a-b-c) -3$, then the value of 2a − 3b + 4c is
(a) −1 (b) 0
(c) 1 (d) 2

5. If $(3a + 1)^2 + (b-1)^2 + (2c-3)^2 = 0$, then the value of (3a + b + 2c) is equal to
(a) 3 (b) −1
(c) 2 (d) 5

6. The value of expression $\frac{(a-b)^2}{(b-c)(c-a)}+\frac{(b-c)^2}{(a-b)(c-a)}+\frac{(c-a)^2}{(a-b)(b-c)}$
(a) 0 (b) 3
(c) 1/3 (d) 2

7. If $a=\frac{\sqrt{5}+1}{\sqrt{5}-1}$ and $b=\frac{\sqrt{5}-1}{\sqrt{5}+1}$, then the value of $\frac{a^2+ab+b^2}{a^2-ab+b^2}$ is
(a) 3/4 (b) 4/3
(c) 3/5 (d) 5/3

8. If a + b + c + d = 1 , then what is the maximum value of
(1 + a)(1 + b)(1 + c)(1 + d) is
(a) 1 (b) 1/8
(c) 27/64 (d) 625/256

9. If $x=\frac{\sqrt{3}}{2}$, then $\frac{\sqrt{1+x}}{1+\sqrt{1+x}}+\frac{\sqrt{1-x}}{1-\sqrt{1-x}}$ is equal to
(a) 1 (b) $2/\sqrt{3}$
(c) $2-\sqrt{3}$ (d) 2

10. If $a^2 + b^2 + c^2 + 3 = 2(a-b-c)$, then the value of 2a−b+c is
(a) 3 (b) 4
(c) 0 (d) 2

11. If $\frac{4x-3}{x}+\frac{4y-3}{y}+\frac{4z-3}{z}=0$, then the value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is
(a) 9 (b) 3
(c) 4 (d) 6

12. If $\frac{xy}{x+y}=a$, $\frac{xz}{x+z}=b$ and $\frac{yz}{y+z}=c$, where a, b, c are all non-zero numbers , then x is equal to
(a) $\frac{2abc}{ab+bc-ac}$ (b) $\frac{2abc}{ab+ac-bc}$
(c) $\frac{2abc}{ac+bc-ab}$ (d) $\frac{2abc}{ab+bc+ac}$

13. If $a+\frac{1}{b}=1$ and $b+\frac{1}{c}=1$ then $c+\frac{1}{a}$ is equal to
(a) 0 (b) 1/2
(c) 1 (d) 2

14. If $x^2+y^2+\frac{1}{x^2}+\frac{1}{y^2}=4$, then the value of $x^2 + y^2$ is
(a) 2 (b) 4
(c) 8 (d) 16

15. If $x^2 = y + z$, $y^2 = z + x$, $z^2 = x + y$ then the value of $\frac{1}{1+x}+\frac{1}{1+y}+\frac{1}{1+z}$ is
(a) −1 (b) 1
(c) 2 (d) 4

16. If $a^2 + b^2 = 2$ and $c^2 + d^2 = 1$, then the value of $(ad-bc)^2 + (ac + bd)^2$ is
(a) 4/9 (b) 1/2

(c) 1 (b) 2

17. If a , b , c are real and $a^2 + b^2 + c^2 + 3 = 2(a + b + c)$, then the value of a + b + c is
(a) 2 (b) 3
(c) 4 (d) 5

18. If $x = \frac{4ab}{a+b}(a \neq b)$, then the value of $\frac{x+2a}{x-2a} + \frac{x+2b}{x-2b}$.
(a) a (b) b
(c) 2ab (d) 2

19. If $a^2 + b^2 + 2b + 4a + 5 = 0$, then the value of $\frac{a-b}{a+b}$ is
(a) 3 (b) −3
(c) 1/3 (b) −1/3

20. If $x - y = \frac{x+y}{7} = \frac{xy}{4}$, the numerical value of xy is
(a) 4/3 (b) 3/4
(c) 1/4 (d) 1/3

21. If x + y + z = 0, then $\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy}$ = ?
(a) $(xyz)^2$ (b) $x^2 + y^2 + z^2$
(c) 9 (d) 3

22. If a + b + c = 0, then
$\frac{1}{(a+b)(b+c)} + \frac{1}{(a+c)(b+a)} + \frac{1}{(c+a)(c+b)}$ = ?
(a) 1 (b) 0
(c) −1 (d) −2

23. If a + b + c = 0, then $\frac{a^2+b^2+c^2}{a^2-bc}$ is
(a) 0 (b) 1
(c) 2 (d) 3

24. If p + q = 10 and pq = 5 , then the numerical value of $\frac{p}{q} + \frac{q}{p}$ will be
(a) 16 (b) 20
(c) 22 (d) 18

25. If $x = 3 + 2\sqrt{2}$ and xy = 1, then the value of $\frac{x^2+3xy+y^2}{x^2-3xy+y^2}$ is
(a) 30/31 (b) 70/31
(c) 35/31 (d) 37/31

26. If $\frac{x}{b+c} = \frac{y}{c+a} = \frac{z}{a+b}$, then
(a) $\frac{x-y}{b-a} = \frac{y-z}{c-b} = \frac{z-x}{a-c}$
(b) $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$
(c) $\frac{x-y}{a} = \frac{y-z}{b} = \frac{z-x}{c}$
(d) None of these

27. If a + b + c = 0, then the value of $\left(\frac{a+b}{c} + \frac{b+c}{a} + \frac{c+a}{b}\right)\left(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}\right)$ is
(a) 8 (b) −3
(c) 9 (d) 0

28. If a, b, c are non-zero, $a + \frac{1}{b} = 1$ and $b + \frac{1}{c} = 1$ then abc is equal to
(a) −1 (b) 3
(c) −3 (d) 1

29. If a + b + c = 2s , then $\frac{(s-a)^2 + (s-b)^2 + (s-c)^2 + s^2}{a^2+b^2+c^2}$ is equal to
(a) $a^2 + b^2 + c^2$ (b) 0
(c) 1 (d) 2

30. If xy + yz + zx = 0 , the $\frac{1}{x^2-yz} + \frac{1}{y^2-zx} + \frac{1}{z^2-xy}$ = ?
(a) 3 (b) 1
(c) x + y + z (d) 0

ANSWER KEY																			
1.	(d)	2.	(a)	3.	(a)	4.	(c)	5.	(a)	6.	(b)	7.	(b)	8.	(d)	9.	(b)	10.	(d)
11.	(c)	12.	(c)	13.	(c)	14.	(a)	15.	(b)	16.	(d)	17.	(b)	18.	(d)	19.	(c)	20.	(a)
21.	(d)	22.	(b)	23.	(c)	24.	(d)	25.	(d)	26.	(a)	27.	(c)	28.	(a)	29.	(c)	30.	(d)

PRACTICE SHEET-3

1. The value of the expression $x^4 - 17x^3 + 17x^2 - 17x + 17$ at $x = 16$ is
(a) 0 (b) 1
(c) 2 (d) 3

2. For what value of k, will the expression $3x^3 - kx^2 + 4x + 16$ be divisible by $\left(x - \frac{k}{2}\right)$?
(a) 4 (b) – 4
(c) 2 (d) 0

3. Which one of the following is the factor of $x^4 + xy^3 + xz^3 + x^3y + y^4 + yz^3$?
(a) $x + y + z$ (b) $x^2 + y^2 + z^2$
(c) $x^3 + y^3 + z^3$ (d) $x^2 + y^2$

4. If one of the two factors of an expression which is the difference of two cubes, is $(x^4 + x^2y + y^2)$, then what is the other factor?
(a) $x + y$ (b) $x - y$
(c) $x^2 + y$ (d) $x^2 - y$

5. Which one of the following is one of the factors of $x^2(y - z) + y^2(z - x) - z(xy - yz - zx)$?
(a) $x-y$ (b) $x + y - z$
(c) $x - y - z$ (d) $x + y + z$

6. If $(3x^3 - 2x^2y - 13xy^2 + 10y^3)$ is divided by $(x - 2y)$, then what is the remainder?
(a) 0 (b) $y + 5$
(c) $y + 1$ (d) $y^2 + 3$

7. If $(x + y + z = 0)$, then what is $(x + y)(y + z)(z + x)$ equal to?
(a) $-xyz$ (b) $x^2 + y^2 + z^2$
(c) $x^3 + y^3 + z^3 + 3xyz$ (d) xyz

8. If $(x^3 + 5x^2 + 10k)$ leaves remainder $-2x$ when divided by $(x^2 + 2)$, then what is the value of k?
(a) –2 (b) –1
(c) 1 (d) 2

9. If $(5x^2 + 14x + 2)^2 - (4x^2 - 5x + 7)^2$ is divided by $(x^2 + x + 1)$, what is the remainder?
(a) –1 (b) 0
(c) 1 (d) 2

10. What is/are the factor(s) of $(x^{29} - x^{24} + x^{13} - 1)$?
(a) Only $(x - 1)$
(b) Only $(x + 1)$
(c) $(x - 1)$ and $(x + 1)$
(d) Neither $(x - 1)$ nor $(x + 1)$

11. What is the value of the polynomial $r(x)$, so that $f(x) = g(x)\,q(x) + r(x)$ and deg $r(x)$ < deg $g(x)$, where $f(x) = x^2 + 1$ and $g(x) = x + 1$?
(a) 1 (b) –1
(c) 2 (d) –2

12. If a is a rational number such that $(x - a)$ is a factor of the polynomial $x^3 - 3x^2 - 3x + 9$, then
(a) a can be any integer
(b) a is an integer dividing 9
(c) a cannot be an integer
(d) a can take three values

13. If $x^2 - 11x + a$ and $x^2 - 14x + 2a$ have a common factor, then what are the values of a?
(a) 0, 7 (b) 5, 20
(c) 0, 24 (d) 1, 3

14. Which one of the following is a factor of $2x^3 - 3x^2 - 11x + 6$?
(a) $x + 1$ (b) $x - 1$
(c) $x + 2$ (d) $x - 2$

15. What should be subtracted from $27x^3 - 9x^2 - 6x - 5$ to make it exactly divisible by $(3x - 1)$?
(a) –5 (b) –7
(c) 5 (d) 7

16. What is $x(y-z)(y+z)+y(z-x)(z+x)+z(x-y)(x+y)$ equal to?
(a) $(x+y)(y+z)(z+x)$
(b) $(x-y)(x-z)(z-y)$
(c) $(x+y)(z-y)(x-z)$
(d) $(y-x)(z-y)(x-z)$

17. If the remainder of the polynomial $a_0 + a_1x + a_2x^2 + ... + a_nx^n$ when divided by $(x-1)$ is 1, then which one of the following is correct?
(a) $a_0 + a_2 + ... = a_1 + a_3 + ...$
(b) $a_0 + a_2 + ... = 1 + a_1 + a_3 + ...$
(c) $1 + a_0 + a_2 + ... = -(a_1 + a_3 + ...)$
(d) $1 - a_0 - a_2 - ..... = a_1 + a_3 + ...$

18. When $(x^3 - 2x^2 + px - q)$ is divided by $(x^2 - 2x - 3)$, the remainder is $(x-6)$. What are the values of p, q respectively?
(a) –2, –6 (b) 2, –6
(c) –2, 6 (d) 2, 6

19. If $x = 1 + \sqrt{2}$, then what is the value of $x^4 - 4x^3 + 4x^2$?
(a) –1 (b) 0
(c) 1 (d) 2

20. When $x^{40} + 2$ is divided by $x^4 + 1$, what is the remainder?
(a) 1 (b) 2
(c) 3 (d) 4

21. If the expression $px^3 + 3x^2 - 3$ and $2x^3 - 5x + p$ when divided by $x-4$ leave the same remainder, then what is the value of p?
(a) –1 (b) 1
(c) –2 (d) 2

22. If u, v and w are real numbers such that $u^3 - 8v^3 - 27w^3 = 18uvw$, then which one of the following is correct?
(a) $u - v + w = 0$ (b) $u = -v = -w$
(c) $u - 2v = 3w$ (d) $u + 2v = -3w$

23. $x^4 + 4y^4$ is divisible by which one of the following?
(a) $(x^2 + 2xy + 2y^2)$ (b) $(x^2 + 2y^2)$
(c) $(x^2 - 2y^2)$ (d) None of these

24. What are the factors of $x^2 + 4y^2 + 4y - 4xy - 2x - 8$?
(a) $(x - 2y - 4)$ and $(x - 2y + 2)$
(b) $(x - y + 2)$ and $(x - 4y + 4)$
(c) $(x - y + 2)$ and $(x - 4y - 4)$
(d) $(x + 2y - 4)$ and $(x + 2y + 2)$

25. If the expression $(px^3 + x^2 - 2x - q)$ is divisible by $(x-1)$ and $(x+1)$, what are the values of p and q, respectively?
(a) 2, –1 (b) –2, 1
(c) –2, –1 (d) 2, 1

26. If $x^5 - 9x^2 + 12x - 14$ is divisible by $(x-3)$, then what is the remainder?
(a) 0 (b) 1
(c) 56 (d) 184

27. If $3x^3 - 2x^2y - 13xy^2 + 10y^3$ is divided by $x - 2y$, then what is the remainder?
(a) 0 (b) x
(c) $y + 5$ (d) $x - 3$

28. What is the remainder when $(x^{11} + 1)$ is divided by $(x+1)$?
(a) 0 (b) 2
(c) 11 (d) 12

29. If $(x-3)$ is a factor of $(x^2 + 4px - 11p)$, then what is the value of p?
(a) –9 (b) –3
(c) –1 (d) 1

30 What is the value of k that $(2x-1)$ may be a factor of $4x^4 - (k-1)x^3 + kx^2 - 6x + 1$?
(a) 8 (b) 9
(c) 12 (d) 13

ANSWER KEY																			
1.	(b)	2.	(b)	3.	(c)	4.	(d)	5.	(c)	6.	(a)	7.	(a)	8.	(c)	9.	(b)	10.	(a)
11.	(c)	12.	(d)	13.	(c)	14.	(c)	15.	(b)	16.	(b)	17.	(d)	18.	(c)	19.	(c)	20.	(c)
21.	(b)	22.	(c)	23.	(a)	24.	(a)	25.	(d)	26.	(d)	27.	(a)	28.	(a)	29.	(a)	30.	(d)

CDS PYQ

1. The sum of two numbers is 7 and the sum of their squares is 25. The product of the two numbers is:
[CDS 2013 (1)]
(a) 6 (b) 10
(c) 12 (d) 15

2. One of the factors of the polynomial $x^4 - 7x^3 + 5x^2 - 6x + 81$ is:
[CDS 2013 (1)]
(a) x+2 (b) x–2
(c) x+3 (d) x–3

3. $(a+1)^4 - a^4$ is divisible by:
[CDS 2013 (1)]
(a) $-2a^2 + 2a - 1$ (b) $2a^3 - 2a - 1$
(c) $2a^3 - 2a + 1$ (d) $2a^2 + 2a + 1$

4. The factor(s) of 5px – 10qy + 2rpx = 4qry is/are:
[CDS 2013 (1)]
(a) Only (5 + 2r)
(b) Only (px – 2qy)
(c) Both (5 + 2r) and (px – 2qy)
(d) Neither (5 + 2r) nor (px – 2qy)

5. If the expression $x^3 + 3x^2 + 4x + p$ contains (x + 6) as a factor, then the value of p is:
[CDS 2013 (1)]
(a) 132 (b) 141
(c) 144 (d) 151

6. What is the value of :
$$\frac{725\times725\times725+371\times371\times371}{725\times725-725\times371+371\times371}?$$
[CDS 2013 (1)]
(a) 9610 (b) 1960
(c) 1096 (d) 1016

7. If $x + \frac{1}{x} = 2$, then what is value of $x - \frac{1}{x}$?
[CDS 2013 (1)]
(a) 0 (b) 1
(c) 2 (d) –2

8. Consider the following statements:
I. x+3 is the factor of $x^3 + 2x^2 + 3x + 8$
II. x–2 is the factor of $x^3 + 2x^2 + 3x + 8$
Which of the statements given above is/are correct?
[CDS 2013 (2)]
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

9. What is $\frac{(x^2+y^2)(x-y)-(x-y)^3}{x^2y-xy^2}$ equal to?
[CDS 2013 (2)]
(a) 1 (b) 2
(c) 4 (d) –2

10. If the expression $x^3 + 3x^2 + 4x + k$ has a factor x + 5, then what is the value of k?
[CDS 2013 (2)]
(a) –70 (b) 70
(c) 48 (d) –48

11. The quantity which must be added to $(1-x)(1+x^2)$ to obtain x^3 is:
[CDS 2013 (2)]
(a) $2x^3 + 3x^2 + x + 1$ (b) $2x^3 + x^2 + x - 1$
(c) $2x^3 - x^2 + x - 1$ (d) $-x^2 + x - 1$

12. If $(49)^2 - (25)^2 = 37x$, then what is x equal to?
[CDS 2013 (2)]
(a) 64 (b) 74
(c) 48 (d) 42

13. For what values of k will:
$4x^5 + 9x^4 - 7x^3 - 5x^2 - 4kx + 3k^2$ contain x–1 as a factor?
[CDS 2013 (2)]
(a) $3, -\frac{1}{2}$ (b) 3, –1
(c) $0, \frac{1}{3}$ (d) $1, \frac{1}{3}$

14. $x(y^2 - z^2) + y(z^2 - x^2) + z(x^2 - y^2)$ is divisible by:
[CDS 2013 (2)]

(a) Only (y–z)
(b) Only (z–x)
(c) Both (y–z) and (z–x)
(d) Neither (y–z) nor (z–x)

15. If $\frac{x}{2}+\frac{y}{3}=4$ and $\frac{2}{x}+\frac{3}{y}=1$`, then what is x + y equal to?

[CDS 2014 (1)]

(a) 11 (b) 10
(c) 9 (d) 8

16. $x^3 + 6x^2 + 11x + 6$ is divisible by:

[CDS 2014 (1)]

(a) Only (x+1) (b) Only (x+2)
(c) Only (x+3) (d) All of these

17. What should be added to be x(x+a) (x+2a) (x+3a), so that, the sum be a perfect square?

[CDS 2014 (1)]

(a) $9a^2$ (b) $4a^2$
(c) a^4 (d) None of these

18. If $3x^4 - 2x^3 + 3x^2 - 2x + 3$ is divided by (3x + 2), then the remainder is:

[CDS 2014 (1)]

(a) 0 (b) $\frac{185}{27}$
(c) $\frac{181}{25}$ (d) $\frac{3}{4}$

19. The expression $2x^3 + x^2 - 2x - 1$ is divisible by:

[CDS 2014 (1)]

(a) x+2 (b) 2x+1
(c) x–2 (d) 2x–1

20. If $\left(x^2+\frac{1}{x^2}\right)=\frac{17}{4}$, then what is $\left(x^3-\frac{1}{x^3}\right)$ equal to?

[CDS 2014 (1)]

(a) $\frac{75}{16}$ (b) $\frac{63}{8}$
(c) $\frac{95}{8}$ (d) None of these

21. If x + y = 5, y + z = 10 and z + x = 15, then which one of the following is correct?

[CDS 2014 (1)]

(a) x > x > y (b) z < y > x
(c) z > y > z (d) x > z > y

22. What is the remainder when $x^5 - 5x^2 + 125$ is divided by x+5?

[CDS 2014 (2)]

(a) 0 (b) 125
(c) –3125 (d) 3125

23. Consider the following statements:
1.(a–b–c) is one of the factor of $3abc + b^3 + c^3 - a^3$
2.(b+c–1) is one of the factor of $3bc + b^3 + c^3 - 1$
Which of the above statement is/are correct?

[CDS 2014 (2)]

(a) Only 1 (b) Only 2
(c) Both 1and 2 (d) Neither 1 nor 2

24. If ax + by – 2 = 0 and abxy = 1, where a ≠ 0, b ≠ 0, then what is $(a^2x + b^2y)$ equal to?

[CDS 2014 (2)]

(a) a + b (b) 2ab
(c) $a^3 + b^3$ (d) $a^4 + b^4$

25. For what value of k is (x–5) a factor of $x^3 - 3x^2 + kx - 10$?

[CDS 2015 (1)]

(a) –8 (b) 4
(c) 2 (d) 1

26. The expression $x^3q^2 - x^3pt + 4x^2pt - 4x^2q^2 + 3xq^2 - 3xpt$ is divisible by:

[CDS 2015 (1)]

(a) (x–1) only
(b) (x–3) only
(c) Both (x–1) and (x–3)
(d) Neither (x–1) nor (x–3)

27. If x + y + z = 0 then $x^3 + y^3 + z^3 + 3xyz$ is equal to:

[CDS 2015 (1)]

(a) 0 (b) 6 xyz
(c) 12 xyz (d) xyz

28. What are the possible solution for x of the equation $x^{\sqrt{x}}=\sqrt[n]{x^x}$, where n and x are positive.

[CDS 2015 (1)]

(a) 0, n (b) 1, n^2
(c) n, n (d) 1, n

29. The square root of $\frac{(0.75)^3}{1-0.75}+\left[0.75+(0.75)^2+1\right]$

[CDS 2015 (1)]

(a) 1 (b) 2
(c) 3 (d) 4

30. Out of 532 saving accounts held in a post office, 218 accounts have deposits over Rs 10,000 each. Further, in 302 accounts, the first or sole depositors are men, of which the deposits exceed Rs 10,000 in 102 accounts. In how many accounts the first or sole depositors are women and the deposits are up to Rs 10,000 only?

[CDS 2015 (1)]

(a) 116
(b) 114
(c) 100
(d)Cannot be determined from the given data

31. Consider the following in respect of the equation $y = \frac{\sqrt{(x-1)^2}}{x-1}$.

1. y = 1 if x > 1
2. y = – 1 if x < 1
3. y exists for all values of x

Which of the above statements is/are correct?

[CDS 2015 (2)]

(a) 1 only (b) 2 only
(c) 1 and 2 only (d) 1, 2 and 3

32. If a, b and c satisfy the equation $x^3 - 3x^2 + 2x + 1 = 0$ then what is the value of $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$?

[CDS 2015 (2)]

(a) $-\frac{1}{2}$ (b) 2
(c) –2 (d) $\frac{1}{2}$

33. Which one of the following is correct?

[CDS 2015 (2)]

(a) (x+2) is a factor of $x^4-6x^3+12x^2-24x+32$
(b) (x+2) is a factor of $x^4-6x^3-12x^2+24x-32$
(c) (x–2) is a factor of $x^4-6x^3+12x^2-24x+32$
(d) (x–2) is a factor of $x^4+6x^3-12x^2+24x-32$

34. If x = $\frac{91}{216}$, then the value of 3 – $\frac{1}{(1-x)^{1/3}}$ is:

[CDS 2015 (2)]

(a) $\frac{9}{5}$ (b) $\frac{a^2-1}{a}=5$
(c) $\frac{4}{9}$ (d) $\frac{4}{5}$

35. If $\frac{p}{x}+\frac{q}{y}=m$ and $\frac{q}{x}+\frac{p}{y}=n$, then what is $\frac{x}{y}$ equal to?

[CDS 2016 (1)]

(a) $\frac{np+mq}{mp+nq}$ (b) $\frac{np+mq}{mp-nq}$
(c) $\frac{np-mq}{mp-nq}$ (d) $\frac{np-mq}{mp+nq}$

36. If $a^2 - by - cz = 0$, $ax - b^2 + cz = 0$ and $ax + by - c^2 = 0$, then the value of $\frac{x}{a+x}+\frac{y}{b+y}+\frac{z}{c+z}$ will be:

[CDS 2016 (1)]

(a) a + b + c (b) 3
(c) 1 (d) 0

37. If (s–a) + (s–b) + (s–c) = s, then the value of $\frac{(s-a)^2+(s-b)^2+(s-c)^2+s^2}{a^2+b^2+c^2}$, will be:

[CDS 2016 (1)]

(a) 3 (b) 1
(c) 0 (d) –1

38. If the polynomial $x^6 + px^5 + qx^4 - x^2 - x - 3$ is divisible by $(x^4 - 1)$, then the value of $p^2 + q^2$ is:

[CDS 2016 (1)]

(a) 1 (b) 9
(c) 10 (d) 13

39. Let m be a non-zero integer and n be a positive integer. Let R be the remainder obtained on dividing the polynomial $x^n + m^n$ by (x – m). Then:

[CDS 2016 (1)]

(a) R is a non zero even integer
(b) R is odd, if m is odd
(c) R = s^2 for some integer s, if n is even
(d) R = t^3 for some integer t, if 3 divides n

40. If $x = \frac{\sqrt{a+2b}+\sqrt{a-2b}}{\sqrt{a+2b}-\sqrt{a-2b}}$ then $bx^2 - ax + b$ is equal to (given that $b \neq 0$):

[CDS 2016 (1)]

(a) 0 (b) 1
(c) ab (d) 2ab

41. If $a^3 = 117 + b^3$ and $a = 3 + b$, then the value of a + b is (given that a > 0 and b > 0)

[CDS 2016 (1)]

(a) 7 (b) 9
(c) 11 (d) 13

42. For what value of k is (x+1) a factor of $x^3 + kx^2 - x + 2$?

[CDS 2016 (1)]

(a) 4 (b) 3
(c) 1 (d) –2

43. If $\sqrt{\frac{x}{y}} = \frac{10}{3} - \sqrt{\frac{y}{x}}$ and x – y = 8, then value of xy is equal to:

[CDS 2016 (1)]

(a) 36 (b) 24
(c) 16 (d) 9

44. If x = $2^{\frac{1}{3}} + 2^{-\frac{1}{3}}$, then the value of $2x^3 - 6x - 5$ is equal to:

[CDS 2016 (1)]

(a) 0 (b) 1
(c) 2 (d) 3

45. If $x^2 = y + z$, $y^2 = z + x$ and $z^2 = x + y$, then what is the value of $\frac{1}{x+1}+\frac{1}{y+1}+\frac{1}{z+1}$?

[CDS 2016 (2)]

(a) –1 (b) 1
(c) 2 (d) 4

46. If 4x + 3a = 0, then what is the value of $\frac{x^2+ax+a^2}{x^3-a^3} - \frac{x^2-ax+a^2}{x^3+a^3}$?

[CDS 2016 (2)]

(a) $-\frac{4}{7a}$ (b) $\frac{7}{a}$
(c) $-\frac{32}{7a}$ (d) $\frac{24}{7a}$

47. If 2p + 3q = 12 and $4p^2 + 4pq - 3q^2 = 126$, then what is the value of p + 2q?

[CDS 2016 (2)]

(a) 5 (b) $\frac{21}{4}$
(c) $\frac{25}{4}$ (d) $\frac{99}{16}$

48. What is the square root of $\frac{(0.35)^2+0.70+1}{2.25}+0.19$?

[CDS 2017 (1)]

(a) 1 (b) 2
(c) 3 (d) 4

49. If x = $\frac{\sqrt{a+b}-\sqrt{a-b}}{\sqrt{a+b}+\sqrt{a-b}}$, then what is $bx^2 - 2ax + b$ equal to ($b \neq 0$)?

[CDS 2017 (1)]

(a) 0 (b) 1
(c) ab (a) 2ab

50. If $x = t^{\frac{1}{t-1}}$ and $y = t^{\frac{t}{t-1}}$, t > 0, t ≠ 0, then what is the relation between x and y?

[CDS 2017 (1)]

(a) $y^x = x^{1/y}$ (b) $x^{1/y} = y^{1/x}$
(c) $x^y = y^x$ (d) $x^y = y^{1/x}$

51. If A : B = 3 : 4, then what is the value of the expression $\left(\frac{3A^2+4B}{3A-4B^2}\right)$?

[CDS 2017 (1)]

(a) $\frac{43}{55}$

(b) $-\frac{43}{55}$

(c) $\frac{47}{55}$

(d) Cannot be determined

52. If $x = 2 + 2^{2/3} + 2^{1/3}$, then what is the value of $x^3 - 6x^2 + 6x$?

[CDS 2017 (1)]

(a) 3 (b) 2
(c) 1 (d) 0

53. If $\sqrt{\frac{x}{y}} = \frac{24}{5} + \sqrt{\frac{y}{x}}$ and x + y = 26, then what is the value of xy?

[CDS 2017 (1)]

(a) 5 (b) 15
(c) 25 (d) 30

54. If a + b = 5 and ab = 6, then what is the value of $a^3 + b^3$?

[CDS 2017 (1)]

(a) 35 (b) 40
(c) 90 (d) 125

55. What is the value of $\frac{(443+547)^2 + (443-547)^2}{443\times443+547\times547}$?

[CDS 2017 (1)]

(a) 0 (b) 1
(c) 2 (d) 3

56. If $9^x\ 3^y = 2187$ and $2^{3x}\ 2^{2y} - 4^{xy} = 0$, then what can be the value of (x + y)?

[CDS 2017 (1)]

(a) 1 (b) 3
(c) 5 (d) 7

57. What are the factors of $x^3 + 4x^2 - 11x - 30$?

[CDS 2017 (2)]

(a) (x–2), (x+3) and (x+5)
(b) (x+2), (x+3) and (x–5)
(c) (x+2), (x–3) and (x+5)
(d) (x+2), (x–3) and (x–5)

58. If x = 111.....1 (20 digits), y = 333...... 3 (10 digits) and z = 222.....2 (10 digits), then what is $\frac{x-y^2}{z}$ equal to?

[CDS 2017 (2)]

(a) 1/2 (b) 1
(c) 2 (d) 3

59. If $5x^3 + 5x^2 - 6x + 9$ is divided by (x + 3), then the remainder is:

[CDS 2017 (2)]

(a) 135 (b) –135
(c) 63 (d) –63

60. The quotient of $8x^3 - y^3$ when divided by $2xy + 4x^2 + y^2$ is:

[CDS 2017 (2)]

(a) 2x + y (b) x+ 2y
(c) 2x–y (d) 4x – y

61. If (x + 2) is a common factor of $x^2 + ax + b$ and $x^2 + bx + a$, then the ratio a : b is equal to:

[CDS 2017 (2)]

(a) 1 (b) 2
(c) 3 (d) 4

62. Let $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \_\_\_\_\_ + a_{n-1} x + a_n$, where $a_0, a_1, a_2, \_\_\_\_\_, a_n$ are real numbers.
If f(x) is divided by (ax–b), then the remainder is:

[CDS 2017 (2)]

(a) $f\left(\frac{b}{a}\right)$ (b) $f\left(-\frac{b}{a}\right)$
(c) $f\left(\frac{a}{b}\right)$ (d) $f\left(-\frac{a}{b}\right)$

63. The product of the polynomials (x + 2), (x – 2), $(x^3 - 2x^2 + 4x - 8)$ and $(x^3 + 2x^2 + 4x + 8)$ is:

[CDS 2017 (2)]

(a) $x^8 - 256$ (b) $(x^4 - 16)^2$
(c) $(x^4 + 16)^2$ (d) $(x^2 - 4)^4$

64. The factor of x (x +2) (x + 3) (x + 5) – 72 are

[CDS 2017 (2)]

(a) x,(x+3), (x+4) and (x–6)
(b) (x–1), (x+6) and $(x^2 - 2x - 12)$
(c) (x–1), (x+6) and $(x^2+5x+12)$
(d) (x– 1), (x–6) and $(x^2 - 5x - 12)$

65. a, b, c, d are non- zero integers such that (ab) divides (cd). If a and c co-prime, then which one of the following is correct?

[CDS 2017 (2)]

(a) a is a factor of c
(b) a is a factor of b
(c) a is a factor of d
(d) d is the factor of a

66. If ab + bc + ca = 0, then what is the value of $\frac{a^2}{a^2-bc} + \frac{b^2}{b^2-ca}+\frac{c^2}{c^2-ab}$?

[CDS 2017 (2)]

(a) 3 (b) 0
(c) 1 (d) –1

67. What is $\frac{(x-y)(x-z)(z-x)}{(x-y)^3+(y-z)^3+(z-x)^3}$ equal to?

[CDS 2017 (2)]

(a) $-\frac{1}{3}$ (b) $\frac{1}{3}$
(c) 3 (d) –3

68. Let f(x) and g(x) be two polynomials (with real coefficients) having degrees 3 and 4 respectively. What is the degree of f(x)g(x)?

[CDS 2017 (2)]

(a) 12 (b) 7
(c) 4 (d) 3

69. The non-zero solution of the equation $\frac{a-x^2}{bx}-\frac{b-x}{c}=\frac{c-x}{b}-\frac{b-x^2}{cx}$, where b ≠ 0, c ≠ 0 is:

[CDS 2017 (2)]

(a) $\frac{b^2+ac}{b^2+c^2}$ (b) $\frac{b^2-ac}{b^2-c^2}$
(c) $\frac{b^2-ac}{b^2+c^2}$ (d) $\frac{b^2+ac}{b^2-c^2}$

70. What is the value of $\frac{1}{1+x^{b-a}+x^{c-a}}+\frac{1}{1+x^{a-b}+x^{c-b}}+\frac{1}{1+x^{a-c}+x^{b-c}}$?

[CDS 2018 (1)]

(a) –1 (b) 0
(c) 1 (d) 3

71. If a + b = 2c, then what is the value of $\frac{a}{a-c}+\frac{c}{b-c}$?

[CDS 2018 (1)]

(a) –1 (b) 0
(c) 1 (d) 2

72. If x + y + z = 0, then what is $(y + z - x)^3 + (z + x - y)^3 + (x + y - z)^3$ equal to?

[CDS 2018 (1)]

(a) $(x + y + z)^3$
(b) 3(x + y) (y + z) (z + x)
(c) 24xyz
(d) –24xyz

73. If $\frac{b}{y}+\frac{z}{c}=1$ and $\frac{c}{z}+\frac{x}{a}=1$, then what is $\frac{ab+xy}{bx}$ equal to:

[CDS 2018 (1)]

(a) 1 (b) 2
(c) 0 (d) –1

74. If $\frac{a^2-1}{a}=5$, then what is the value of $\frac{a^6-1}{a^3}$?

[CDS 2018 (1)]

(a) 125 (b) –125
(c) 140 (d) –140

75. If (x + 3) is a factor of $x^3 + 3x^2 + 4x + k$, then what is the value of k?

[CDS 2018 (1)]

(a) 12 (b) 24
(c) 36 (d) 72

76. Which one of the following is a zero of the polynomial $3x^3 + 4x^2 - 7$?

[CDS 2018 (1)]

(a) 0 (b) 1
(c) 2 (d) –1

77. The remainder when $3x^3 + kx^2 + 5x - 6$ is divided by (x + 1) is –7. What is the value of k?

[CDS 2018 (1)]

(a) –14 (b) 14
(c) –7 (d) 7

78. If f(x) and g(x) are polynomials of degree p and q respectively, then the degree of {f(x) ± g(x)} (if it is non-zero) is:

[CDS 2018 (1)]

(a) Greater than min (p,q)
(b) Greater than max. (p,q)
(c) Less than or equal to max. (p,q)
(d) Equal to min (p,q)

79. If $x = y^{1/a}$, $y = z^{1/b}$ and $z = x^{1/c}$ where x ≠ 1, y ≠ 1, z ≠ 1, then what is the value of abc?

[CDS 2018 (1)]

(a) –1 (b) 1
(c) 0 (d) 3

80. If 2b = a + c and $y^2 = xz$, then what is $x^{b-c}\, y^{c-a}\, z^{a-b}$ equal to?

[CDS 2018 (1)]

(a) 3 (b) 2
(c) 1 (d) –1

81. The sum of a number and its square is 20. Then the number is

[CDS 2018 (1)]

(a) –5 or 4 (b) 2 or 3
(c) –5 only (d) 5 or –4

82. If $x = y^a$, $y = z^b$ and $z = x^c$, then the value of abc is:

[CDS 2018 (2)]

(a) 1 (b) 2
(c) –1 (d) 0

83. If $x = 2 + 2^{2/3} + 2^{1/3}$, then the value of the expression $x^3 - 6x^2 + 6x$ will be:

[CDS 2018 (2)]

(a) 2 (b) 1
(c) 0 (d) –2

84. If $x^6 + \frac{1}{x^6} = k\left(x^2 + \frac{1}{x^2}\right)$, then k is equal to:

[CDS 2018 (2)]

(a) $\left(x^2 - 1 + \frac{1}{x^2}\right)$ (b) $\left(x^4 - 1 + \frac{1}{x^4}\right)$
(c) $\left(x^4 + 1 + \frac{1}{x^4}\right)$ (d) $\left(x^4 - 1 - \frac{1}{x^4}\right)$

85. If $a^x = b^y = c^z$ and abc = 1, then the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ will be equal to:

[CDS 2018 (2)]

(a) –1 (b) 0
(c) 1 (d) 3

86. If $a = xy^{p-1}$, $b = yz^{q-1}$, $c = zx^{r-1}$,, then $a^{q-r}\, b^{r-p}\, c^{p-q}$ is equal to:

[CDS 2018 (2)]

(a) abc
(b) xyz
(c) 0
(d) none of the above

87. The remainder when $3x^3 - 2x^2y - 13xy^2 + 10y^3$ is divided by (x–2y) is equal to:

[CDS 2018 (2)]

(a) zero (b) y
(c) y–5 (d) y+3

88. If ab + bc + ca = 0, then the value of
$$\frac{(b^2-ca)(c^2-ab)+(a^2-bc)(c^2-ab)+(a^2-bc)(b^2-ca)}{(a^2-bc)(b^2-ca)(c^2-ab)}$$
is:

[CDS 2018 (2)]

(a) –1 (b) 0
(c) 1 (d) 2

89. If α, β and γ are the zeros of the polynomial $f(x) = ax^3 + bx^2 + cx + d$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to:

[CDS 2018 (2)]

(a) $\frac{b^2 - ac}{a^2}$ (b) $\frac{b^2 - 2ac}{a}$
(c) $\frac{b^2 - 2ac}{b^2}$ (d) $\frac{b^2 - 2ac}{a^2}$

90. For x > 0, what is minimum value of $x + \frac{x+2}{2x}$?

[CDS 2019 (1)]

(a) 1
(b) 2
(c) $2\frac{1}{2}$
(d) cannot be determined

91. If $\frac{1+px}{1-px}\sqrt{\frac{1-qx}{1+qx}}$ = 1, then what are the non – solutions of x?

[CDS 2019 (1)]

(a) $\pm\frac{1}{p}\sqrt{\frac{2p-q}{q}}$, 2p ≠q (b) $\pm\frac{1}{pq}\sqrt{p-q}$, p≠q
(c) $\pm\frac{p}{q}\sqrt{p-q}$, p≠q (d) $\pm\frac{q}{p}\sqrt{2p-q}$, 2p≠q

92. For $x = \frac{4\sqrt{6}}{\sqrt{2}+\sqrt{3}}$, what is the value of $\frac{x+2\sqrt{2}}{x-2\sqrt{2}} + \frac{x+2\sqrt{3}}{x-2\sqrt{3}}$?

[CDS 2019 (1)]

(a) 1 (b) $\sqrt{2}$
(c) $\sqrt{3}$ (d) 2

93. For any two numbers a and b, $\sqrt{(a-b)^2}+\sqrt{(b-a)^2}$ is:

[CDS 2019 (1)]

(a) Always zero
(b) Never zero
(c) Positive only if $a \neq b$
(d) Positive if and only if $a > b$

94. If $\frac{a}{b+c}+\frac{b}{c+a}=\frac{c}{a+b}$, then which one of the following statements is correct?

[CDS 2019 (1)]

(a) Each fraction is equal to 1 or –1
(b) Each fraction is equal to 1/2 or 1
(c) Each fraction is equal to 1/2 or –1
(d) Each fraction is equal to 1/2 only

95. If $3^x = 4^y = 12^z$, then z is equal to:

[CDS 2019 (1)]

(a) xy (b) x + y
(c) $\frac{xy}{x+y}$ (d) 4x + 3y

96. Given that the polynomial $(x^2 + ax + b)$ leaves the same remainder when divided by (x–1) or (x + 1). What are the values of a and b respectively?

[CDS 2019 (1)]

(a) 4 and 0 (b) 0 and 3
(c) 3 and 0 (d) 0 and any integer

97. β If $a^x = b^y = c^z$ and $b^2 = ac$, then what is $\frac{1}{x}+\frac{1}{z}$ equal to?

[CDS 2019 (1)]

(a) $\frac{1}{y}$ (b) $-\frac{1}{y}$
(c) $\frac{2}{y}$ (d) $-\frac{2}{y}$

98. What is $\frac{(x-y)^3+(y-z)^3+(z-x)^3}{3(x-y)(y-z)(z-x)}$ equal to?

[CDS 2019 (1 & 2)]

(a) 1 (b) 0
(c) 1/3 (d) 3

99. For what value of k can the expression $x^3 + kx^2 - 7x + 6$ be resolved into three linear factors?

[CDS 2019 (2)]

(a) 0 (b) 1
(c) 2 (d) 3

100. The quotient when $x^4 - x^2 + 7x + 5$ is divided by (x + 2) is $ax^3 + bx^2 + cx + d$. What are the values of a, b, c and d respectively?

[CDS 2019 (2)]

(a) 1, –2, 3, 1 (b) –1, 2, 3, 1
(c) 1, –2, –3, –1 (d) –1, 2, –3, –1

101. Let a and b two positive real numbers such that $a\sqrt{a}+b\sqrt{b}=32$ and $a\sqrt{a}+b\sqrt{b}=31$. What is the value of $\frac{5(a+b)}{7}$?

[CDS 2019 (2)]

(a) 5
(b) 7
(c) 9
(d) Cannot be determined

102. If x = $\frac{1+\sqrt{3}}{2}$ and y = x^3, the y satisfies which one of the following equations?

[CDS 2019 (2)]

(a) $8y^2 - 20y - 1 = 0$ (b) $8y^2 + 20y - 1 = 0$
(c) $8y^2 - 20y + 1 = 0$ (d) $8y^2 - 20y + 1 = 0$

103. Give y is inversely proportional to $\sqrt{x}$, and x = 36, when y = 36. What is the value of x when y = 54?

[CDS 2019 (2)]

(a) 54 (b) 27
(c) 16 (d) 8

104. What are the values of p and q respectively, if (x–1) and (x + 2) divide the polynomial $x^3 + 4x^2 + px + q$?

[CDS 2020 (1)]

(a) 1,–6 (b) 2,–6
(c) 1,6 (d) 2,6

105. What is the point on the xy-plane satisfying 5x + 2y = 7xy and 10x + 3y = 8xy?

[CDS 2020 (1)]

(a) $\left(-1,\frac{1}{6}\right)$ (b) $\left(\frac{1}{6},-1\right)$

(c) $\left(1, \frac{1}{6}\right)$ (d) $\left(-\frac{1}{6}, -1\right)$

106. Which one of the following is a set of solutions of the equation $x^{\sqrt{x}} = \sqrt[n]{x^x}$ if n is a positive integer?

[CDS 2020 (1)]

(a) $\{1, n^2\}$ (b) $\{1, \sqrt{n}\}$
(c) $\{1, n\}$ (d) $\{n, n^2\}$

107. If $5^{x+1} - 5^{x-1} = 600$, then what is the value of 10^{2x}?

[CDS 2020 (1)]

(a) 1 (b) 1000
(c) 100000 (d) 1000000

108. If f(x) is divided by $(x - \alpha)(x - \beta)$ where $\alpha \neq \beta$, then what is the remainder?

[CDS 2020 (1)]

(a) $\frac{(x-\alpha)f(\alpha)-(x-\beta)f(\beta)}{\alpha-\beta}$

(b) $\frac{(x-\alpha)f(\beta)-(x-\beta)f(\alpha)}{\alpha-\beta}$

(c) $\frac{(x-\beta)f(\alpha)-(x-\alpha)f(\beta)}{\alpha-\beta}$

(d) $\frac{(x-\beta)f(\beta)-(x-\alpha)f(\alpha)}{\alpha-\beta}$

109. If the area of a square is $2401x^4 + 196x^2 + 4$, then what is its side length?

[CDS 2020 (1)]

(a) $49x^2 + 3x + 2$ (b) $49x^2 - 3x + 2$
(c) $49x^2 + 2$ (d) $59x^2 + 2$

110. If x varies as yz, then y varies inversely as:

[CDS 2020 (1)]

(a)xz (b) $\frac{x}{z}$
(c) $\frac{z}{x}$ (d) $\frac{1}{(xz)}$

111. If $\left(x^8 + \frac{1}{x^8}\right) = 47$, what is the value of $\left(x^6 + \frac{1}{x^6}\right)$?

[CDS 2020 (1)]

(a) 36 (b) 27
(c) 18 (d) 9

112. If $(x^2 - 1)$ is a factor of $ax^4 + bx^3 + cx^2 + dx + e$, then which one of the following is correct?

[CDS 2020 (1)]

(a) a + b + c = d + e (b) a + b + e = c + d
(c) b + c + d = a + e (d) a + c + e = b + d

113. If $I = a^2 + b^2 + c^2$, where a and b are consecutive integers and c = ab, then I is:

[CDS 2020 (2)]

(a)An even number and it is not a square of an integer
(b)An odd number and it is not a square of an integer
(c)Square of an even integer
(d)Square of an odd integer

114. $(x^n - a^n)$ is divisible by (x–a), where $x \neq a$, for every:

[CDS 2020 (2)]

(a) Natural number n
(b) Even natural number n only
(c) Odd natural number n only
(d) Prime number n only

115. If x varies as y, then which of the following is/are correct?

1. $x^2 + y^2$ varies as $x^2 - y^2$
2. $\frac{x}{y^2}$ varies inversely as y
3. $\sqrt[n]{x^2y}$ varies as $\sqrt[2n]{x^4y^2}$

Select the correct answer using the code given below:

[CDS 2020 (2)]

(a) 1 and 2 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

116. If ab + xy – xb = 0 and bc + yz – cy = 0, then what is $\frac{x}{a} + \frac{c}{z}$ equal to?

[CDS 2020 (2)]

(a) $\frac{y}{b}$ (b) $\frac{b}{y}$
(c) 1 (d) 0

117. If (p + 2) (2q – 1) = 2pq – 10 and (p – 2) (2q – 1) = 2pq – 10, then what is pq equal to?
[CDS 2020 (2)]
(a) –10 (b) –5
(c) 5 (d) 10

118. What is the value of :
$$\frac{a^2+ac}{a^2c-c^3}-\frac{a^2-c^2}{a^2c+2ac^2+c^3}-\frac{2c}{a^2-c^2}+\frac{3}{a+c}?$$
[CDS 2020 (2)]
(a) 0 (b) 1
(c) $\frac{ac}{a^2+c^2}$ (d) $\frac{6}{a+c}$

119. What is the square root of $4x^4 + 8x^3 - 4x + 1$?
[CDS 2020 (2)]
(a) $2x^2 - 2x - 1$ (b) $2x^2 - x - 1$
(c) $2x^2 + 2x + 1$ (d) $2x^2 + 2x - 1$

120. If $\frac{x}{b+c}=\frac{y}{c+a}=\frac{z}{b-a}$, then which one of the following is correct?
[CDS 2020 (2)]
(a) x + y + z = 0 (b) x – y – z = 0
(c) x + y – z = 0 (d) x + 2y + 3z = 0

121. $1-x-x^n+x^{n+1}$, where n is a natural number, is divisible by:
[CDS 2020 (2)]
(a) $(1+x)^2$ (b) $(1-x)^2$
(c) $1-2x-x^2$ (d) $1+2x-x^2$

122. $4x^3 + 12x^2 - x - 3$ is divisible by:
[CDS 2021 (1)]
(a) (2x + 1) only
(b) (2x –1) only
(c) Both (2x +1) and (2x –1)
(d) Neither (2x +1) nor (2x –1)

123. If $\frac{x}{a}+\frac{y}{b}=a+b$ and $\frac{x}{a^2}+\frac{y}{b^2}=2$, then what is $\frac{x}{a^2}-\frac{y}{b^2}$ equal to?
[CDS 2021 (1)]
(a) –2 (b) –1
(c) 0 (d) 1

124. How many pairs of (x, y) can be chosen from the set {2, 3, 6, 8, 9} such that $\frac{x}{y}+\frac{y}{x}=2$, where $x \neq y$?
[CDS 2021 (1)]
(a) Zero (b) One
(c) Two (d) Three

125. How many terms are there in following product?
$(a_1 + a_2 + a_3)(b_1 + b_2 + b_3 + b_4)$
$(c_1 + c_2 + c_3 + c_4 + c_5)$
[CDS 2021 (1)]
(a) 15 (b) 30
(c) 45 (d) 60

126. What is the remainder when $27^{27}-15^{27}$ is divided by 6?
[CDS 2021 (1)]
(a) 0 (b) 1
(c) 3 (d) 4

127. If a + b + c = 0 then which of the following are correct?
1. $a^3 + b^3 + c^3 = 3abc$
2. $a^2 + b^2 + c^2 = -2(ab + bc + ca)$
3. $a^3 + b^3 + c^3 = -3ab(a + b)$
Select the correct answer using the code given below:
[CDS 2021 (1)]
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

128. If $p=\frac{\sqrt{3q+2}+\sqrt{3q-2}}{\sqrt{3q+2}-\sqrt{3q-2}}$, then what is the value of $p^2 - 3pq + 2$?
[CDS 2021 (1)]
(a) 0 (b) 1
(c) 2 (d) 3

129. What is the value of x, if: $\frac{b+\sqrt{b^2-2bx}}{b-\sqrt{b^2-2bx}}=a$?
[CDS 2021 (1)]
(a) $\frac{ab}{(a+b)}$ (b) $\frac{2ab}{(a+1)}$
(c) $\frac{2ab}{(a+1)^2}$ (d) $\frac{ab}{(a+b)^2}$

130. For how many real values of k is $6kx^2 + 12kx - 24x + 16$ a perfect square for every integer x?

[CDS 2021 (1)]

(a) Zero (b) One
(c) Two (d) Four

131. If $x + \frac{1}{x} = \frac{5}{2}$, then what is $x^4 - \frac{1}{x^4}$ equal to?

[CDS 2021 (1)]

(a) $\frac{195}{16}$ (b) $\frac{255}{16}$
(c) $\frac{625}{16}$ (d) 0

132. The expression $\frac{(x^3-1)(x^2-9x+14)}{(x^2+x+1)(x^2-8x+7)}$ simplifies to:

[CDS 2021 (1)]

(a) $(x-1)$ (b) $(x-2)$
(c) $(x-7)$ (d) $(x+2)$

133. What should be added to $\frac{1}{(x-2)(x-4)}$ to get $\frac{2x-5}{(x^2-5x+6)(x-4)}$?

[CDS 2021 (1)]

(a) $\frac{1}{(x^2-7x+12)}$ (b) $\frac{1}{(x^2+7x+12)}$
(c) $\frac{1}{(x^2-7x-12)}$ (d) $\frac{1}{(x^2+7x-12)}$

134. Consider the following statements:
1.If x is directly proportional to z and y is directly proportional to z, then $(x^2 - y^2)$ is directly proportional to z^2 ?
2.If x is inversely proportional to z and y is inversely proportional to z, then (xy) is inversely proportional to z^2 ?
Which of the above statements is/are correct?

[CDS 2021 (1)]

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

135. What is $\frac{8x}{1-x^4} - \frac{4x}{x^2+1} + \frac{x+1}{x-1} - \frac{x-1}{x+1}$ equal to?

[CDS 2021 (1)]

(a) 0 (b) 1
(c) 2 (d) 4

136. If $x(x-1)(x-2)(x-3) + 1 = k^2$, then which one of the following is a possible expression for k?

[CDS 2021 (1)]

(a) $x^2 - 3x + 1$ (b) $x^2 - 3x - 1$
(c) $x^2 + 3x - 1$ (d) $x^2 - 2x - 1$

137. What is
$$\frac{1}{bc(a-b)(a-c)} + \frac{1}{ca(b-c)(b-a)} + \frac{1}{ab(c-a)(c-b)}$$
equal to:

[CDS 2021 (1)]

(a) a+ b + c (c) 3
(c) ab + bc + ca (d) 0

138. If n is any natural number, then $5^{2n} - 1$ is always divisible by how many natural numbers?

[CDS 2021 (1)]

(a)One (b) Four
(c) Six (d) Eight

139. Which one of the following is a factor of the polynomial $(x-1)(x-2)(x-4) - 90$?

[CDS 2021 (2)]

(a) x + 14 (b) x – 14
(c) x – 6 (d) x – 7

140. What is the sum of linear factors (in x and y) of the expression $2x^2 + xy - 3y^2$?

[CDS 2021 (2)]

(a) 2x–3y (b) 3x –2y
(c) 3x + 2y (d) 2x + 3y

141. If $\alpha + \beta + \gamma = \alpha\beta + \beta\gamma + \gamma\alpha$, then what is $(1-\alpha)(1-\beta)(1-\gamma)$ equal to?

[CDS 2021 (2)]

(a) $1-\alpha\beta\gamma$ (b) $1+\alpha\beta\gamma$
(c) $\alpha^2 + \beta^2 + \gamma^2$ (d) $(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$

142. If $96 - 64a^3 + \frac{8}{a^6} - \frac{48}{a^3} - t^3 = 0$, then what is $a^2t + 4a^3$ equal to?

[CDS 2021 (2)]

(a) 0 (b) 1
(c) 2 (d) 3

143. If A + B = $\frac{x^2-8}{x+2}$ and $A-B=\frac{-x^2+2x+4}{x+2}$; then what is B equal to?

[CDS 2021 (2)]

(a) $\frac{x^2-4}{x^2+4x+4}$ (b) $\frac{x^2-4}{x^2-4x+4}$
(c) $\frac{2x^2-7x+3}{2x-1}$ (d) $\frac{2x^2+7x-3}{2x-1}$

144. What is $\frac{x^4}{(x^2-y^2)(x^2-z^2)}+\frac{y^4}{(y^2-x^2)(y^2-x^2)}+\frac{z^4}{(z^2-x^2)(z^2-y^2)}$ equal to?

[CDS 2021 (2)]

(a) –1 (b) 0
(c) 1 (d) $x^2 + y^2 + z^2$

145. If $(2ab - b^2) : (6a^2 - ab) = 1 : 6$, then what is $(a + b) : (a - b)$ equal to?

[CDS 2021 (2)]

(a) 3 only (b) 5 only
(c) –3 or 3 (d) –5 or 5

146. If $\frac{x}{b+c-a}=\frac{y}{b-c-a}=\frac{z}{a+b-c}=k$, then what is:
$x^2 + y^2 + z^2 - 2xy - 2yz + 2zx$ equal to?

[CDS 2021 (2)]

(a) $k^2 (a^2 + b^2 + c^2)$ (b) $k^2 (a^2 - b^2 + c^2)$
(c) $k^2 (a + b + c)^2$ (d) $k^2 (a - b + c)^2$

147. Consider the following inequalities:

1. $\frac{a^2-b^2}{a^2+b^2}>\frac{a-b}{a+b}$ where a > b > 0
2. $\frac{a^3+b^3}{a^2+b^2}>\frac{a^2+b^2}{a+b}$ only when a > b > 0

Which of the above is/are correct?

[CDS 2021 (2)]

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

148. If $\frac{ay-bx}{c}=\frac{cx-az}{b}=\frac{bz-cy}{a}$, then which of the following is/are correct?

1. $\frac{x}{a}=\frac{y}{b}$
2. $\frac{x+y+z}{a+b+c}=\frac{z}{c}$

Select the correct answer using the code given below:

[CDS 2021 (2)]

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

149. If $2x - 3y - 7 = 0$, then what is the value of $8x^3 - 36x^2y + 54xy^2 - 27y^3 - 340$?

[CDS 2021 (2)]

(a) –1 (b) 0
(c) 1 (d) 3

150. Consider a question and the two statements:
Question:
Is $\frac{x^6+y^6}{x^4+y^4}$ always greater than $\frac{x^4+y^4}{x^2+y^2}$ $(x \neq y \neq 0)$?
Statement-I: x > y
Statement-II: $x^2 + y^2 > 2xy$
Which one of the following is correct in respect of the question and the statements?

[CDS-2022-1]

(a)Statement-I alone is required to answer the question
(b)Statement-II alone is required to answer the question
(c)Both Statements-I & II are required to answer the question.
(d)Both Statement-I nor statement-II is required to answer the question.

151. If $x+\frac{1}{x}=\frac{5}{2}$ then what is the value of the following? $\frac{5x}{7x^2-3x+7}$

[CDS-2022-1]

(a) $\frac{3}{7}$ (b) $\frac{5}{12}$
(c) $\frac{3}{14}$ (d) $\frac{10}{29}$

152. If a + b = 2, $\frac{1}{a}+\frac{1}{b}$ = 2 then what is the value of $a^3 + b^3$?

[CDS-2022-1]

(a) 2 (b) 4
(c) 6 (d) 8

153. If $43^x + 47^y = (2021)^2$, $x \neq 0$, $y \neq 0$, then what is the value of the following?

$$\frac{4xy + x + y}{2xy - x - y}$$

[CDS-2022-1]

(a) 5 (b) 15
(c) 25 (d) 45

154. If $\frac{x - y}{x\sqrt{y} + y\sqrt{x}} = \frac{1}{\sqrt{x}}$; $(x > 0, y > 0)$, then what is the value of $\frac{x}{y}$?

[CDS-2022-1]

(a) 1 (b) 2
(c) 4 (d) 8

155. If $x = 7 + 4\sqrt{3}$, then what is the value of $\sqrt{x} + \frac{1}{\sqrt{x}}$?

[CDS-2022-1]

(a) 1 (b) 2
(c) 3 (d) 4

156. What is the value of the following?

$$\frac{(5.4)^3 - 0.064}{(5.4)^2 + 2.16 + 0.16}$$

[CDS-2022-1]

(a) 4 (b) 4.4
(c) 5 (d) 5.4

157. If $\left(x + \sqrt{1 + x^2}\right)\left(y + \sqrt{1 + y^2}\right) = 1$, where x and y are real numbers, then what is the value of $(x + y)^2$?

[CDS-2022-1]

(a) 0 (b) 1
(c) 4 (d) 9

158. If x = 9999, then what is the value of the following?

$$\frac{4x^3 - x}{(2x + 1)(6x - 3)}$$

[CDS-2022-1]

(a) 1111 (b) 2222
(c) 3333 (d) 6666

159. Consider a question and two statements:
Question: $a^2 + b^2 + c^2 - ab - bc - ca$ (a, b, c are distinct real numbers), always positive?
Statement-I: $a > b > c$
Statement-II: $a + b + c = 0$
Which one of the following is correct in respect of the question and the statements?

[CDS-2022-1]

(a)Statement-I alone is required to answer the question
(b)Statement-II alone is required to answer the question
(c)Both Statements-I & II are required to answer the question.
(d)Both Statement-I nor statement-II is required to answer the question.

160. If $x^2 = 17x + y$ and $y^2 = x + 17y$, $x \neq y$, then what is the value of $\sqrt{x^2 + y^2 + 1}$?

[CDS-2022-1]

(a) 17 (b) 19
(c) 23 (d) 27

Consider the following for the next two (02) items that follow:

Consider $6x^2 - 25x + \frac{6}{x^2} + \frac{25}{x} + 12 = 0$

161. What is one of the possible values of $x - \frac{1}{x}$?

[CDS 2022 (II)]

(a) $\frac{1}{2}$ (b) $\frac{3}{2}$
(c) 2 (d) $\frac{5}{2}$

162. What is one of the possible values of $x^2 + \frac{1}{x^2}$?

[CDS 2022 (II)]

(a) 6 (b) $\frac{62}{9}$
(c) 8 (d) $\frac{82}{9}$

163. If bc + cd = 2bd and a + c = 2b, then which one of the following is correct?

[CDS 2022 (II)]

(a) ab – cd = 0 (b) ac – bd = 0
(c) ad – bc = 0 (d) ad + bc = 0

164. If x = b + c, y = c + a, z = a + b, then what is $(x + y + z)^3 - 24xyz$ equal to?

[CDS 2022 (II)]

(a) $a^3 + b^3 + c^3$ (b) $2(a^3 + b^3 + c^3)$
(c) $8(a^3 + b^3 + c^3)$ (d)none of the above

165. If p is the difference between a number and its reciprocal and q is the difference between the square of the same number and the square of its reciprocal, then what is $p^4 + 4p^2$ equal to?

[CDS 2022 (II)]

(a) 4q (b) 8q
(c) $4q^2$ (d) q^2

166. If $x = 2 + 2^{1/2}$, then what is the value of $x^4 + 16x^{-4}$?

[CDS 2022 (II)]

(a) 152 (b) 144
(c) 136 (d) 132

167. Consider the following statements:

1. $x^4 (y-z) + y^4 (z-x) + z^4(x-y)$ is positive if $x > y > z$.
2. $x^4 (y-z) + y^4 (z-x) + z^4(x-y)$ is negative if $x < y < z$.

Which of the above statements is/are correct?

[CDS 2022 (II)]

(a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

168. What is $\frac{x^6 - 24x^4 + 144x^2}{\left(x^2 + 4\sqrt{3}x + 12\right)\left(x - 2\sqrt{3}\right)^2}$ equal to?

[CDS 2022 (II)]

(a) x^2 (b) x^2-2
(c) $x^2 + 2\sqrt{3}$ (d) $x^2 - 2\sqrt{3}$

169. If $\frac{a+b}{b+c} = \frac{c+d}{d+a}$ where $a \neq c$, then which one of the following is correct?

[CDS 2022 (II)]

(a) $a + b = c + d$ (b) $a + c = b + d$
(c) $a - b - c + d = 0$ (d) $a + b + c + d = 0$

170. If $(a^3 + b^3)$ is proportional to $(a^2 - b^2)$, then $(a^2 - ab + b^2)$ is proportional to:

[CDS 2022 (II)]

(a) $(a - b)$ (b) $(a + b)$
(c) $(a + ab + b)$ (d) $(a^3 - b^2)$

171. If $\frac{\sqrt{x+20}+\sqrt{x-1}}{\sqrt{x+20}-\sqrt{x-1}} = \frac{7}{3}$, then what is the value of $\sqrt{(x+20)(x-1)}$?

[CDS 2022 (II)]

(a) 8 (b) 9
(c) 10 (d) 12

172. Consider the following statements:

1. $(ab + bc + ca)$ is a factor of $a^2 (b - c)^3 + b^2 (c - a)^3 + c^2 (a - b)^3$
2. $(a + b + c)$ is a factor of $a^2 (b - c)^3 + b^2 (c - a)^3 + c^2 (a - b)^3$

Which of the statements given above is/are correct?

[CDS 2022 (II)]

(a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

173. What is

$$\frac{1}{x(x-y)(x-z)} + \frac{1}{y(y-z)(y-x)} + \frac{1}{z(z-x)(z-y)}$$

equal to?

[CDS 2022 (II)]

(a) 0 (b) 1
(c) $\frac{1}{xyz}$ (d) $-\frac{1}{xyz}$

174. How many of the following values of x would satisfy the equation $qx^2-2px+q = 0$?

1. $x = \frac{\sqrt{p+q}+\sqrt{p-q}}{\sqrt{p+q}-\sqrt{p-q}}$, where $p > q$

2. $x = \frac{p+\sqrt{p^2-q^2}}{q}$, where $p > q$

3. $x = \frac{\sqrt{p+q}}{\sqrt{p-q}}$, where $p > q$

Select the correct answer using the code given below:

[CDS 2022 (II)]

(a) only one value (b) only two values
(c) all three values (d) none

175. Consider the question and two statements given below:

Question: Is $(x^n + y^n)$ divisible by $(x + y)$?
Statement-1: n is a natural number.
Statement-2: n is an even natural number.

Which one of the following is correct in respect of the question and the statements?

[CDS 2022 (II)]

(a)Statement-1 alone is sufficient to answer the question
(b)Statement-2 alone is sufficient to answer the question
(c)Both statement-1 and statement-2 are sufficient to answer the question
(d)Both statement-1 and statement-2 are not sufficient to answer the question

176. Consider the question and two statements given below:

Let x and y be two real numbers.
Question: Is $xy > 0$?
Statement-1: $x^8 y^9 < 0$.

Statement-2: $x^9 y^{10} < 0$
Which one of the following is correct in respect of the question and the statements?
[CDS 2022 (II)]
(a)Statement-1 alone is sufficient to answer the question
(b)Statement-2 alone is sufficient to answer the question
(c)Both statement-1 and statement-2 are sufficient to answer the question
(d)Both statement-1 and statement-2 are not sufficient to answer the question

177. Consider the following statements in respect of the polynomial $1 - x - x^n + x^{n+1}$, where n is a natural number:
1.It is divisible by $1 - 2x + x^2$
2.It is divisible by $1 - x^n$
Which of the statements given above is/are correct?
[CDS – 2023 (1)]
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

178. If $x = a + b + \frac{(a-b)^2}{4a+4b}$ and $y = \frac{a+b}{4} + \frac{ab}{a+b}$, then what is the value of $(x-a)^2 - (y-b)^2$?
[CDS – 2023 (1)]
(a) a^2 (b) b^2
(c) ab (d) a^2b^2

179. If $x = \frac{\sqrt{3}+1}{\sqrt{3}-1}$ and $y = \frac{\sqrt{3}-1}{\sqrt{3}+1}$, then what is the value $x^3 - y^3$?
[CDS – 2023 (1)]
(a) 60 (b) $45\sqrt{3}$
(c) $30\sqrt{3}$ (d) 90

180. If $p = \frac{a^2}{(b-c)(c-a)}, q = \frac{b^2}{(c-b)(a-b)}, r = \frac{c^2}{(a-c)(b-c)}$, then what is $(p + q + r)^2$ equal to?
[CDS-2023 (2)]
(a) 9 (b) 4
(c) 1 (d) 0

181. Which one of the following is a factor of $a^2 - b^2 - c^2 + 2bc + a + b - c$?
[CDS-2023 (2)]
(a) a + b + c + 1 (b) a – b – c + 1
(c) a + b + c –1 (d) a – b + c + 1

182. If 2s = a + b + c, then what is $s^2 + (s-a)(s-b) + (s-b)(s-c) + (s-c)(s-a)$ equal to?
[CDS-2023 (2)]
(a) $(a+b+c)^2$ (b) ab + bc + ca
(c) 2(ab+bc+ca) (d) 3(ab+bc+ca)

183. The cube root of x varies inversely as the square root of y. x = 8 when y = 3. What is the value of x when y = $\sqrt[3]{3}$?
[CDS-2023 (2)]
(a) 18 (b) 21
(c) 24 (d) 27

184. What is $\frac{x^2-y^2-z^2-2yz}{x^2+y^2-z^2+2xy} + \frac{x^2-y^2-z^2-2yz}{x^2-y^2+z^2-2xz}$ equal to?
[CDS-2023 (2)]
(a) $\frac{x}{x+y-z}$ (b) $\frac{y+z}{x+y-z}$
(c) $\frac{2x}{x+y-z}$ (d) $\frac{2y+2z}{x+y-z}$

185. If a, b, c, x, y, z are real numbers such that $(a + b + c)^2 - 3(ab + bc + ca) + 3(x^2 + y^2 + z^2) = 0$, then which one of the following is correct?
[CDS-2023 (2)]
(a) a = b = c, x = y = z ≠ 0
(b) a = b = c = 0, x = y = z
(c) a = b = c, x = y = z = 0
(d) a ≠ b ≠ c, x = y = z = 0

Consider the following for the next two items that follow:
Mark option (a) if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

Mark option (b) if the question can be answered by using either statements alone.

Mark option (c) if the question canbe answered by using both the statements together, but cannot be answered using either statement alone.

Mark option (d) if the question cannot be answered even by using both the statements together.

186. Question: Are x, y, z equal, where x, y, z are real numbers?
Statement I:
$x^2 + y^2 + z^2 - xy - yz - zx = 0$
Statement II:
$x^3 + y^3 + z^3 - 3xyz = 0$

187. Question: What is the ratio x : y : z equal to if x, y, z ≠ 0?

Statement I:

$\frac{x+z}{y}=\frac{z}{x}$

Statement II:

$\frac{z-y}{x}=\frac{x}{z}$

188. If $a^2-bc=\alpha$, $b^2-ac=\beta$, $c^2-ab=\gamma$, then what is $\frac{a\alpha+b\beta+c\gamma}{(a+b+c)(\alpha+\beta+\gamma)}$ equal to?

[CDS-2023 (2)]

(a) $a+b-c$ (b) $a-b+c$
(c) $-a+b+c$ (d) 1

189. If $(x-1)^3$ is a factor of $x^4+\alpha x^3+\beta x^2+\gamma x-1$, then the other factor will be:

[CDS-2023 (2)]

(a) $x+1$ (b) $x-3$
(c) $x+2$ (d) x

190. If $\frac{2a}{3}=\frac{4b}{5}=\frac{3c}{4}$, then what is the value of $\frac{18}{a}\sqrt{a^2+c^2-b^2}$?

[CDS-2023 (2)]

(a) $3\sqrt{5}$ (b) $\sqrt{355}$
(c) $\sqrt{375}$ (d) $3\sqrt{15}$

191. Consider the following statements:
1.If (a+b) is directly proportional to (a–b), then (a^2+b^2) is directly proportional to ab
2.If a is directly proportional to b, then (a^2-b^2) is directly proportional to ab.
Which of the statements given above is/are correct?

[CDS-2023 (2)]

(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

192. If $(3a+6b+c+2d)\times(3a-6b-c+2d)=(3a-6b+c-2d)\times(3a-c-2d)$, then which one of the following is correct?

[CDS-2023 (2)]

(a) ab = cd (b) ac = bd
(c) ad = bc (d) ad + bc = 0

193. If $x=97+56\sqrt{3}$, then what is the value of $\sqrt[4]{x}+\frac{1}{\sqrt[4]{x}}$?

[CDS-2023 (2)]

(a) 7 (b) 6
(c) 5 (d) 4

194. If $x\left(a-b+\frac{ab}{a-b}\right)=y\left(a+b-\frac{ab}{a+b}\right)$ and $x+y=2a^3$, then what is x–y equal to?

[CDS-2023 (2)]

(a) $-2b^3$ (b) $-2ab^3$
(c) $2b^3$ (d) $2ab^3$

195. Which one of the following is a factor of $3\sqrt{3}x^3+2\sqrt{2}y^3-18xy+6\sqrt{6}$?

[CDS-2023 (2)]

(a) $\sqrt{3}x+\sqrt{2}y-\sqrt{3}$
(b) $\sqrt{3}x+\sqrt{2}y-\sqrt{6}$
(c) $3x^2+2y^2-\sqrt{18}x-\sqrt{12}y-\sqrt{6}xy+6$
(d) $3x^2+2y^2+\sqrt{18}x+\sqrt{12}y-\sqrt{6}xy+6$

196. If $x^n-py^n+qz^n$ is divisible by $x^2+abyz-bzx-axy$, then what is $\frac{p}{a^n}-\frac{q}{b^n}$ equal to?

[CDS-2023 (2)]

(a) –1 (b) 0
(c) 1 (d) 2

Consider the following for the next two (02) items that follow:
Let $p=x^4-y^2z^2$, $q=y^4-z^2x^2$, $r=z^4-x^2y^2$

197. What is $px^2+qy^2+rz^2$ equal to?

[CDS-2023 (2)]

(a) $(x^2+y^2+z^2)(p+q+r)$ (c) $-(x^2+y^2+z^2)(p+q+r)$
(c) $(y^2+z^2+x^2)(p-q-r)$ (d) $(x^2+y^2-z^2)(p-q-r)$

198. What is $x^2(px^2+qy^2+rz^2)+qr-p^2$ equal to?

[CDS-2023 (2)]

(a) 0 (b) 1
(c) p+q+r (d) $x^2+y^2+z^2$

199. Let a, b and c be the sides of a triangle ABC
Question: Is the triangle equilateral?
Statement-I: $a^2+b^2+c^2=(ab+bc+ca)$
Statement-II: $3a^2+3b^2+4c^2=2ab+4bc+4ca$

[CDS-2024 (1)]

(a) If the Question can be answered by one of the Statements alone, but not by the other.
(b) If the Question can be answered by either Statement alone
(c) If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone.
(d) If the Question can be answered even by using both Statements together

200. If P^2 varies as R and Q^2 varies as R, $(P\neq Q)$, then which of the following are correct?
1. P^2+Q^2 varies as R
2. PQ varies as R
3. P^2-Q^2 varies as R

Select the correct answer using the code given below:

[CDS-2024 (1)]

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

201. What is the minimum value of $\left(\frac{a^2+3a+1}{a}\right)\left(\frac{b^2+3b+1}{b}\right)$ for a, b > 0?

[CDS-2024 (1)]

(a) 1 (b) 9
(c) 16 (d) 25

202. Consider the following in respect of the polynomial $x^{4k} + x^{4k+2} + x^{4k+4} + x^{4k+6}$;

1. The remainder is zero when the polynomial is divided by $x^2 + 1$
2. The remainder is zero when the polynomial is divided by $x^4 + 1$

Which of the statements given above is/are correct?

[CDS-2024 (1)]

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

203. If $p = \sqrt[3]{\left(a+\sqrt{a^2+b^3}\right)} + \sqrt[3]{\left(a-\sqrt{a^2+b^2}\right)}$, then what is $p^3 + 3bp$ equal to?

[CDS-2024 (1)]

(a) $-2a$ (b) a
(c) 2a (d) 3a

204. If 2 is a zero of the polynomial $p(x) = x^3 + 3x^2 - 6x - a$, then what is the sum of the squares of the other zeroes of the polynomial?

[CDS-2024 (1)]

(a) 10 (b) 17
(c) 21 (d) 37

205. Suppose $p(x) = x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$ and $q(x) = x^4 + b_3x^3 + b_2x^2 + b_1x + b_0$ are the polynomials. If α, β, γ, δ are zeros of p(x) and α, β, γ, λ are zeros of q(x), then what is $\frac{p(x)-q(x)}{(x-\alpha)(x-\beta)(x-\gamma)}$ equal to?

[CDS-2024 (1)]

(a) $-\lambda + \delta$ (b) $\lambda - \delta$
(c) $\lambda + \delta$ (d) $-\lambda - \delta$

ANSWER KEY

1.	c	**2.**	d	**3.**	d	**4.**	c	**5.**	a	**6.**	c	**7.**	a	**8.**	d	**9.**	b	**10.**	b
11.	c	**12.**	c	**13.**	d	**14.**	c	**15.**	b	**16.**	d	**17.**	c	**18.**	b	**19.**	b	**20.**	b
21.	a	**22.**	c	**23.**	c	**24.**	a	**25.**	a	**26.**	c	**27.**	b	**28.**	d	**29.**	d	**30.**	b
31.	c	**32.**	c	**33.**	c	**34.**	a	**35.**	c	**36.**	c	**37.**	b	**38.**	c	**39.**	c	**40.**	a
41.	a	**42.**	d	**43.**	d	**44.**	a	**45.**	b	**46.**	c	**47.**	a	**48.**	a	**49.**	a	**50.**	c
51.	d	**52.**	b	**53.**	c	**54.**	a	**55.**	c	**56.**	c	**57.**	c	**58.**	b	**59.**	d	**60.**	c
61.	a	**62.**	a	**63.**	b	**64.**	c	**65.**	c	**66.**	c	**67.**	d	**68.**	a	**69.**	b	**70.**	c
71.	c	**72.**	d	**73.**	a	**74.**	c	**75.**	a	**76.**	b	**77.**	d	**78.**	a	**79.**	b	**80.**	c
81.	a	**82.**	a	**83.**	a	**84.**	b	**85.**	b	**86.**	d	**87.**	a	**88.**	b	**89.**	d	**90.**	c
91.	a	**92.**	d	**93.**	c	**94.**	c	**95.**	c	**96.**	d	**97.**	c	**98.**	a	**99.**	a	**100.**	a
101.	c	**102.**	a	**103.**	c	**104.**	a	**105.**	b	**106.**	b	**107.**	d	**108.**	c	**109.**	c	**110.**	c
111.	c	**112.**	d	**113.**	d	**114.**	a	**115.**	b	**116.**	c	**117.**	c	**118.**	d	**119.**	d	**120.**	b
121.	b	**122.**	c	**123.**	c	**124.**	a	**125.**	d	**126.**	a	**127.**	c	**128.**	b	**129.**	c	**130.**	c
131.	b	**132**	b	**133.**	a	**134**	c	**135**	a	**136**	a	**137.**	d	**138.**	d	**139**	d	**140.**	c
141.	a	**142.**	c	**143.**	c	**144.**	c	**145.**	d	**146.**	c	**147.**	c	**148**	c	**149.**	d	**150.**	d
151.	d	**152.**	a	**153.**	a	**154.**	c	**155.**	d	**156.**	c	**157.**	a	**158.**	c	**159.**	a	**160.**	a
161.	b	**162.**	d	**163.**	c	**164.**	c	**165.**	d	**166.**	c	**167.**	c	**168.**	a	**169.**	d	**170.**	a
171.	c	**172.**	a	**173.**	c	**174.**	b	**175.**	b	**176.**	d	**177.**	c	**178.**	b	**179.**	c	**180.**	c
181.	d	**182.**	b	**183.**	c	**184.**	c	**185.**	b	**186.**	a	**187.**	c	**188.**	d	**189.**	a	**190.**	b
191.	c	**192.**	c	**193.**	d	**194.**	c	**195.**	c	**196.**	b	**197.**	a	**198.**	a	**199.**	b	**200.**	d
201.	d	**202.**	c	**203.**	c	**204.**	b	**205.**	b										